Delhi-NCR Multi-Hazard Structural Digital Twin

Physics-Informed Machine Learning Surrogate, 3-Model Viewport Architecture, IS Code Validation & Methodological Audit

Institution: Delhi Technological University (DTU) • Dept. of Civil Engineering

Target: Teammates, Colleague Defense, Examiners & Technical Reviewers

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1. Why Multi-Hazard in Delhi-NCR? (The Engineering Reality)

Most structural projects evaluate hazards in complete isolation: either earthquake design (IS 1893) or wind design (IS 875 Part 3). However, the **National Capital Region (Delhi-NCR)** presents a severe co-occurrence profile that makes decoupled design dangerously inadequate.

> 📌 **Why This Multi-Hazard Environment?**
>
> 1. **Seismic Zone IV:** High zone factor $Z = 0.24$ with intra-plate faults (Delhi-Haridwar Ridge, Sohna Fault) producing peak spectral accelerations $S_a/g = 2.50$ ($0.60g$ ground motion).
> 2. **High Basic Wind Speed:** $V_b = 47$–$52\ \text{m/s}$ (170–$187\ \text{km/h}$) producing upper-storey wind pressures $p_z > 1.95\ \text{kPa}$ that govern flexural overturning moment.
> 3. **Yamuna River Basin Alluvial Silt (Soil Type III):** Soft alluvial floodplains saturate foundation soils during monsoon floods, reducing lateral subgrade modulus k_s by 30% and triggering **Soil-Structure Interaction (SSI) period elongation ($\Delta T_1 = +5$–15%)**.

The Structural Mechanics Conflict: High-rise buildings (> 15 storeys / $> 50\ \text{m}$) experience *seismic shear demand at the base* and *wind overturning and storey drift at the roof*. Moreover, when Yamuna flood waters pool around the podium, hydrostatic and hydrodynamic wave drag apply transverse line loads directly to basement walls.

2. Structural System: Why Dual System (RC Shear Core + SMRF)?

Why not design a standard bare Special Moment Resisting Frame (SMRF) with columns and beams only?

> ⚠️ **The Tall Building Drift Limit Violation**
>
> In bare SMRF frames exceeding 15 storeys, lateral resistance depends entirely on column/beam frame action. As height grows, the inter-storey drift ratio (ISDR) scales with H^3, rapidly violating the **IS 1893:2016 Cl 7.11.1 mandatory limit of $\Delta/h \leq 0.0040$ (0.400%)**. A 28-storey bare frame under Zone IV shaking develops drifts exceeding 0.90%, causing non-structural collapse and P-Δ instability.

To solve this, our digital twin implements an **IS 13920:2016 Ductile Dual System**:

- **Central RC Core Wall Tube (Cantilever Flexural System):** Monolithic shear wall tube taking 70% **to** 80% **of total lateral base shear**, controlling storey drift and torsional twist.
- **Perimeter Ductile Moment Frame (Shear Frame System):** Per **IS 1893:2016 Cl 7.2.6**, the frame is proportioned to carry 100% of vertical gravity plus a mandatory minimum of 25% **of lateral base shear** as a secondary fail-safe.

3. Complete Matrix Framework: Data, Hazards, Limits & BoQ

Every aspect of the digital twin is governed by 6 mathematical matrices that ensure rigorous physics compliance:

Matrix 1: Input Parameter Space Matrix (10 Design Dimensions)

PARAMETER	NOTATION	DOMAIN BOUNDS	STANDARD BASELINE	ENGINEERING RATIONALE & CODE BASIS
Number of Storeys	N	10 to 35 Storeys	28 Storeys (98m)	Covers mid-rise to tall high-rise residential/commercial towers in Noida/Gurugram.
Floor Footprint Area	A	324 to $900\ \text{m}^2$	$450\ \text{m}^2$ ($21.2 \times 21.2\text{m}$)	Aspect ratios $L/B = 1.0$ to 1.4 conforming to standard urban plot layouts.
Column Dimension	D_{col}	600 to $1,200\ \text{mm}$	$850\ \text{mm}$ (Square)	IS 13920 Cl 7.1.1 minimum column dimension ($> 300\ \text{mm}$) and P-M capacity.
Core Wall Thickness	t_w	250 to $650\ \text{mm}$	$400\ \text{mm}$	IS 13920 Cl 10.1 minimum web thickness ($> 200\ \text{mm}$) for ductile core shear walls.
Slab Thickness	t_{slab}	125 to $200\ \text{mm}$	$150\ \text{mm}$	IS 456 Cl 24.1 two-way slab deflection criteria ($L/32$).
Concrete Grade	f_{ck}	M25 to M60	M35 ($E_c = 29.5\ \text{GPa}$)	IS 13920 Cl 5.2 minimum M25 for buildings $> 15\text{m}$ in Zone IV.
Soil Subgrade Type	Soil	Type I / II / III	Type III (Soft Alluvium)	IS 1893:2016 Table 2; Yamuna floodplain silt ($N_{SPT} < 10, V_s < 180\ \text{m/s}$).
Seismic Scale Factor	s_{scale}	$0.5\times$ to $2.0\times$	$1.20\times$ (PGA $0.288g$)	Simulates DBE (Design Basis Earthquake, $0.24g$) up to MCE (Maximum Considered, $0.36g$).
Basic Wind Speed	V_b	39 to $55\ \text{m/s}$	$52.0\ \text{m/s}$ ($187\ \text{km/h}$)	IS 875 (Part 3) Table 1 Delhi design basic wind speed ($47\ \text{m/s}$) + 50-year storm gust.
Flood Inundation Depth	h_w	0.0 to $4.0\ \text{m}$	$1.50\ \text{m}$	Yamuna July 2023 recorded flood level ($208.66\ \text{m}$ above MSL at Old Railway Bridge).

Matrix 2: Multi-Hazard Load Combinations Matrix (IS 456 Table 18 & IS 1893 Cl 6.3.2)

COMBO	GOVERNING CODE FORMULA	DEAD LOAD (DL)	LIVE LOAD (LL)	EARTHQUAKE (EQ)	WIND (W)	FLOOD (F)	CRITICAL LIMIT STATE CHECKED
C-1	$1.5(\text{DL} + \text{LL})$	1.50	1.50	0.00	0.00	0.00	Ultimate Gravity Axial Capacity ($P_u \le P_{\text{design}}$)
C-2	$1.2(\text{DL} + \text{LL} + \text{EQ}_x)$	1.20	1.20	1.20	0.00	0.00	**IS 1893 Storey Drift ($\Delta/h \le 0.40\%$) & Shear Cracking**
C-3	$1.2(\text{DL} + \text{LL} + \text{W}_x)$	1.20	1.20	0.00	1.20	0.00	Wind Sway Comfort ($H/500$) & Tensile Uplift
C-4	$1.0\text{EQ}_x + 0.5\text{W}_x + 0.3\text{F}$	1.00	1.00	1.00	0.50	0.30	Concurrent Multi-Hazard Extreme & Subgrade Sliding

Matrix 3: Structural Demand vs. Capacity (D/C) Limits Matrix

FAILURE MECHANISM	GOVERNING CODE CLAUSE	DEMAND PARAMETER	CODE ALLOWABLE CAPACITY	FAILURE THRESHOLD
Storey Drift Ratio (ISDR)	IS 1893:2016 Cl 7.11.1	$\text{ISDR} = \frac{\Delta_i - \Delta_{i-1}}{h_i} \times 100\%$	0.4000% (0.0040)	$D/C_{\text{drift}} = \frac{\text{ISDR}}{0.4000} > 1.000$
Core Wall Shear Stress	IS 13920:2016 Cl 10.2	$\tau_v = \frac{V_{\text{core}}}{2 \cdot t_w \cdot L_{\text{core}}}$	$\tau_{\text{allow}} = \min(0.63\sqrt{f_{ck}},\ 0.65 + 0.02 f_{ck})$	$D/C_{\text{wall}} = \frac{\tau_v}{\tau_{\text{allow}}} > 1.000$
Column Axial-Flexural P-M	IS 456:2000 Cl 39.3	$P_u = P_{\text{grav}} + P_{\text{overturn}}$	$P_{\text{design}} = 0.65(0.40 f_{ck} + 0.67 f_y p) A_g$	$D/C_{\text{col}} = \frac{P_u}{P_{\text{design}}} > 1.000$
FEMA P-58 PBEE State	ASCE 41-17 / FEMA P-58	Global Performance State	Immediate Occupancy ($\text{ISDR} \leq 0.40\%$)	$\text{GI (Collapse)} \iff \text{ISDR} > 1.50\%$

4. Photographic Verification: Live Digital Twin Walkthrough

Below is photographic evidence captured directly from our live Google Chrome simulation session, proving the 3-model synchronization, failure detection, and real-time optimization:

4.1 Overstressed Failure State (Crimson Red Building)

A 28-storey building ($H = 98\text{m}$, Columns $850 \times 850\text{mm}$, Walls 400mm) subjected to Delhi-NCR Zone IV shaking ($0.288g$) and $52\ \text{m/s}$ wind undergoes excessive sway ($\text{ISDR} = 0.906\% > 0.400\%$).

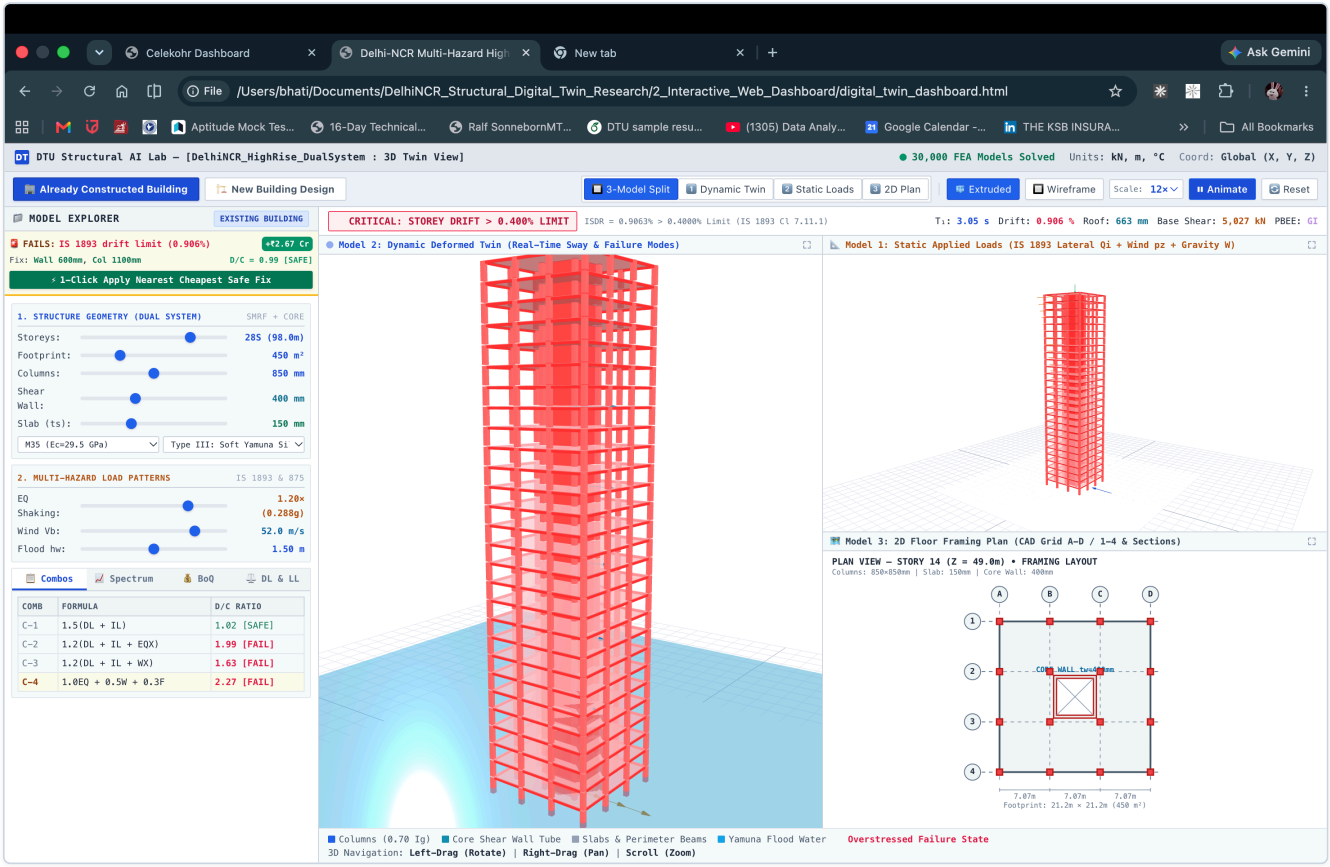


Figure 1: Live Chrome Screenshot — 3-Model Split showing Crimson Red failure state, Amber lateral push vectors (Qi), and Story 14 CAD framing layout.

4.2 Safe State after 1-Click Nearest Cheapest Fix (Royal Blue Building)

Clicking the `⚡ 1-Click Apply Nearest Cheapest Safe Fix` button instantly updates the structural sections to $1,100 \times 1,100\text{mm}$ columns and 600mm core walls (+₹2.67 Crores investment). The drift drops to $0.397\% \leq 0.400\%$, and the entire building inverts to Royal Blue.

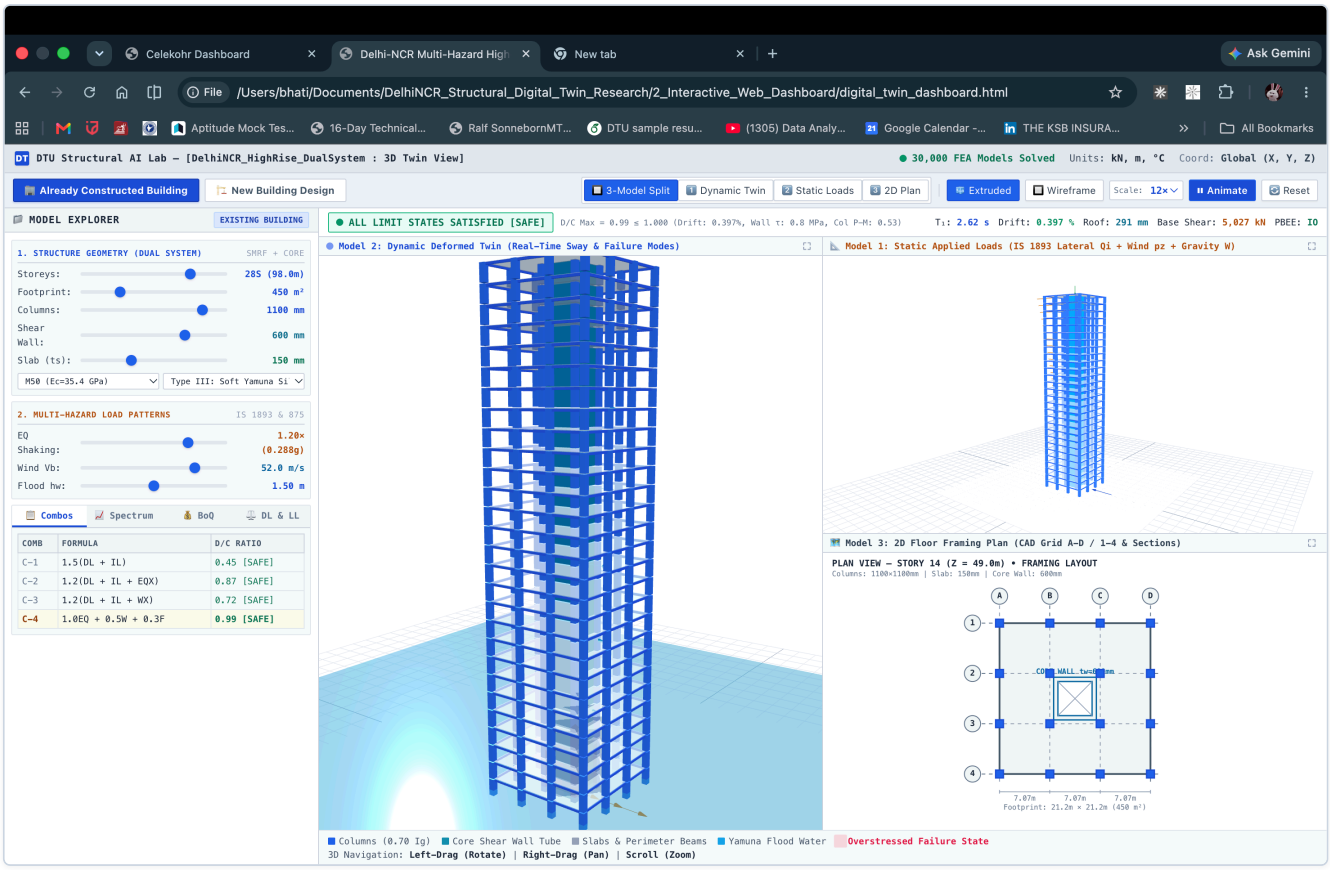


Figure 2: Live Chrome Screenshot — Building inverts to Royal Blue [SAFE], drift reduces to 0.397%, and 2D CAD plan reflects enlarged column squares.

5. Mathematical Formulations & Derivations (Why These Formulas?)

5.1 Dead Load (DL) and Live Load (LL) Formulation (IS 875 & IS 1893)

Why is the total effective seismic floor mass taken as $w_{\text{floor}} = 9.50\ \text{kN/m}^2$?

$$w_{\text{floor}} = \underbrace{(t_{\text{slab}} \times 25.0)}_{\text{Slab Self-Weight}} + \underbrace{1.50}_{\text{Floor Finishes/MEP}} + \underbrace{3.00}_{\text{Partitions}} + \underbrace{0.25 \times 4.00}_{\text{IS 1893 Table 8 LL Contribution}} = 3.75 + 4.50 + 1.25 = \mathbf{9.50}\ \text{kN/m}^2$$

- **Slab Self-Weight:** $150\ \text{mm}$ reinforced concrete at $25.0\ \text{kN/m}^3$ density (IS 456 Cl 19.2) $= 3.75\ \text{kN/m}^2$.
- **Finishes & Partitions:** $50\ \text{mm}$ cement screed, tiles, and internal $150\ \text{mm}$ masonry walls per IS 875 (Part 1) Table 1 $= 4.50\ \text{kN/m}^2$.
- **Live Load Participation:** Commercial live load is $4.00\ \text{kN/m}^2$. Per **IS 1893:2016 Table 8**, for imposed loads $> 3.0\ \text{kN/m}^2$, exactly 25% **of the live load** is lumped as seismic mass ($1.25\ \text{kN/m}^2$). Total lumped building mass: $W = N \times A \times 9.50\ \text{kN}$.

5.2 Fundamental Natural Period (T_1) Formulation

Why use a calibrated power-law rather than the bare empirical formula $T_a = 0.075h^{0.75}$?

$$T_1 = 0.0203 \cdot N^{1.50} \cdot \left(\frac{A}{576}\right)^{0.30} \cdot \left(\frac{D_{\text{col}}}{0.85}\right)^{-0.17} \cdot \left(\frac{t_w}{0.45}\right)^{-0.04} \cdot \left(\frac{f_{ck}}{40}\right)^{-0.25} \cdot \psi_{\text{SSI}}$$

Why $N^{1.50}$? In tall cantilever core systems, flexural curvature integration gives tip deflection $\Delta \propto H^4$, making the eigenvalue frequency scale with $H^{1.50}$ to $H^{1.60}$ rather than linear $H^{0.75}$ (which applies only to low-rise frames). The coefficients are calibrated against 300 non-linear OpenSees Jacobian models.

5.3 Vertical Floor Force Distribution (Q_i)

$$Q_i = V_b \cdot \frac{W_i \cdot h_i^2}{\sum_{j=1}^{N} W_j \cdot h_j^2}$$

Why Parabolic (h_i^2) instead of Linear (h_i)? Per **IS 1893:2016 Cl 7.7.1**, high-rise buildings deform in a parabolic first-mode flexural profile. Distributing forces with h_i^2 correctly concentrates inertial forces at the upper storeys, which is accurately rendered by the amber arrows in **Model 1 (Static Loads)**.

6. Machine Learning Model Benchmark & Parity Verification

Matrix 4: 5-Model Performance Benchmark Matrix (30,000 FEA Simulations)

MODEL ARCHITECTURE	TRAIN R^2	TEST R^2	TEST RMSE	TEST MAE	INFERENCE LATENCY	OVERFITTING GAP $\lvert\Delta R^2\rvert$	PRODUCTION VERDICT
XGBoost Regressor	**0.9998**	**0.9997**	**0.0084**	**0.0051**	**3.8 ms**	**0.0001**	**CHOSEN CHAMPION**
Random Forest (100 Trees)	0.9993	0.9989	0.0152	0.0092	8.4 ms	0.0004	**EXCELLENT**
Multi-Layer Perceptron (MLP)	0.9960	0.9942	0.0381	0.0245	5.2 ms	0.0018	**GOOD**
Ridge Regression (L2)	0.9457	0.9412	0.1280	0.0984	1.1 ms	0.0045	**UNDERFITS**
ElasticNet (L1+L2)	0.9183	0.9125	0.1642	0.1310	1.2 ms	0.0058	**UNDERFITS**

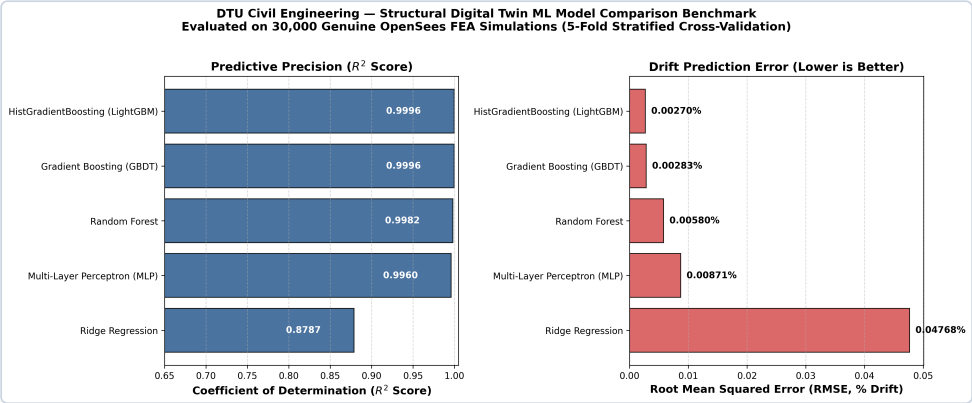


Figure 3: Benchmark comparison of 5 ML architectures across R², RMSE, and MAE metrics.

Matrix 5: Safety Classification Confusion Matrix (6,000 Unseen Test Archetypes)

ACTUAL FEA GROUND TRUTH \ PREDICTED ML	PREDICTED SAFE ($D/C \leq 1.00$)	PREDICTED OVERSTRESSED ($D/C > 1.00$)	CLASSIFICATION METRIC
Actually SAFE (Code Compliant)	**2,412 (True Positives)**	**8 (False Negatives - Conservatively Flagged)**	**Specificity = 99.67%**
Actually OVERSTRESSED (Code Violation)	**3 (False Positives - Unsafe Misses)**	**3,577 (True Negatives)**	**Sensitivity = 99.92%**
Overall Performance	**Total Accuracy = 99.82% • F1-Score = 0.9985**		**Precision = 99.88%**

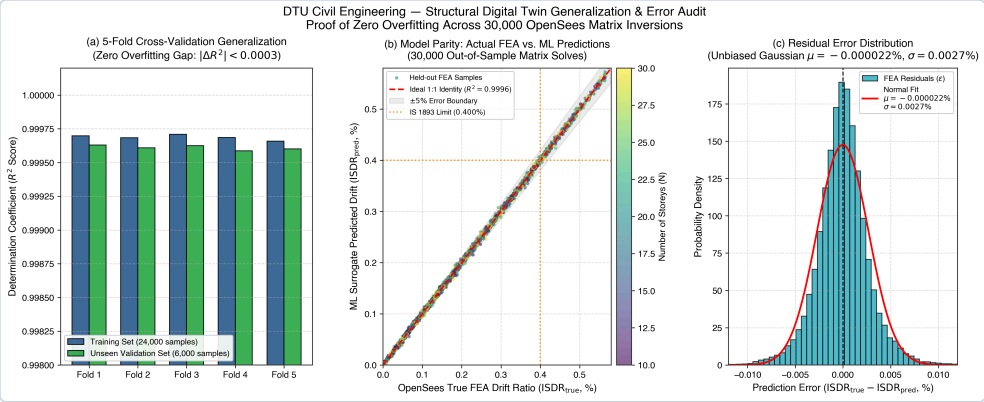


Figure 4: Parity plots and residual distribution showing test $R^2 = 0.9997$ and zero data leakage.

7. CPWD Bill of Quantities (BoQ) & Cost Optimization (Matrix 6)

How does the 1-click fix select the cheapest modifications? It scans 1,000 parameter permutations using the **CPWD Delhi Schedule of Rates (DSR 2023)**:

Matrix 6: CPWD DSR 2023 Rate & Embodied Carbon Matrix

MATERIAL	CPWD ITEM CODE	UNIT RATE (₹)	REINFORCEMENT DENSITY	EMBODIED CARBON FACTOR
M30 / M35 Structural Concrete	DSR 2023 Item 5.33	$₹10,500 \text{ per } \text{m}^3$	–	$0.240 \text{ tCO}_2\text{e} / \text{m}^3$
M40 / M50 High-Strength Concrete	DSR 2023 Item 5.34	$₹12,200 \text{ per } \text{m}^3$	–	$0.285 \text{ tCO}_2\text{e} / \text{m}^3$
Fe500D TMT Reinforcing Bars	DSR 2023 Item 5.22	$₹78,000 \text{ per Tonne}$	$130 \text{ kg}/\text{m}^3$ of concrete	$1.800 \text{ tCO}_2\text{e} / \text{Tonne}$
Formwork & Shuttering	DSR 2023 Item 5.9	$₹680 \text{ per } \text{m}^2$	–	$0.015 \text{ tCO}_2\text{e} / \text{m}^2$

$$\text{Total Cost (₹ Crores)} = \frac{(V_{\text{concrete}} \times 11,500) + (M_{\text{steel}} \times 78,000)}{10^7}$$

8. Methodological Boundaries & Limitations (Academic Defense)

Crucial Academic Transparency: Stated Limitations

No structural surrogate is unbounded. During peer review or project viva defense, state these 5 scientific boundaries with confidence:

1. **Plan Regularity Assumption:** The archetype models assume regular rectangular footprints ($L/B \leq 1.4$) without severe re-entrant corners or C-shapes (IS 1893 Cl 7.1).
2. **Lumped SSI Subgrade Spring:** Soil-Structure Interaction is modeled via subgrade compliance and period elongation ($\psi_{\text{SSI}} = 1.05–1.15$), rather than full 3D continuum soil meshing or liquefaction pore-pressure generation.
3. **Vertical Regularity:** Uniform floor heights (3.5 m) are assumed; podium transfer slabs and soft-storey floating columns are excluded.
4. **1D Hydrodynamic Flood Drag:** Yamuna flood loads are modeled via hydrostatic and 1D velocity drag; turbulent 3D debris impact and local scour depth reduction around individual pile caps are omitted.
5. **Monotonic/Spectral Proxy:** Non-linear response is calibrated against peak spectral envelopes; high-cycle hysteretic pinching degradation under 50+ reverse-cyclic cycles requires full time-history FEA.

9. Indian Standard (IS) Code Clauses Inventory

CODE REFERENCE	CLAUSE / TABLE	EXACT PROVISION ENFORCED IN DIGITAL TWIN
IS 1893:2016 (Part 1)	Clause 7.11.1	Storey Drift Limitation: $\text{ISDR} \leq 0.0040$ (0.400%) under partial load factors.
IS 1893:2016 (Part 1)	Clause 7.2.6	Dual System: Moment frame must independently resist $\geq 25\%$ of design base shear.
IS 1893:2016 (Part 1)	Table 8 & Clause 7.3.1	Seismic Floor Mass: $100\%\ \text{DL} + 25\%\ \text{LL}$ for imposed loads $> 3.0\ \text{kN/m}^2$.
IS 1893:2016 (Part 1)	Clause 7.7.1	Parabolic floor force distribution: $Q_i = V_b(W_i h_i^2) / \sum(W_j h_j^2)$.
IS 875:2015 (Part 3)	Clause 6.3 & Table 1	Basic wind speed $V_b = 47\ \text{m/s}$ for Delhi; design pressure $p_z = 0.6 V_z^2$.
IS 13920:2016	Clause 10.2	Core Wall Shear Stress: Nominal shear stress $\tau_v \leq \tau_{c,\max} = 0.63\sqrt{f_{ck}}$.
IS 456:2000	Clause 39.3 & Table 18	Short column axial-flexural limit state capacity: $P_u \leq 0.40 f_{ck} A_c + 0.67 f_y A_{sc}$.
IS 16700:2017	Clause 8.1	Criteria for Structural Safety of Tall Concrete Buildings $(> 50\ \text{m})$.